QA Small Test Set (5 questions)

JEE Main · Nonofficial practice
Attempt: ae468cf9-6141-4cf8-9257-99d3161c0656
Started: 2026-10-12T07:26:48.160002+00:00
Content version: 1 · Key version: 1
Score version: original

Score: 20 / 20 · Percentage: 100.0% (not a percentile)
Correct: 5 · Incorrect: 0 · Skipped: 0
Dropped: 0 · Bonus-to-all: 0
Attempt rate: 100.0% · Accuracy: 100.0%
Accuracy = correct / attempted; dropped and bonus items excluded.
Penalty marks: 0 · Elapsed: 0 seconds
Rules: +4 correct, -1 wrong, 0 unanswered
Numerical: Exact integer value; signs, Unicode minus and leading zeros accepted; up to 20 digits. No decimals or rounding.

Complete record: all 5 questions in original delivery order.
Skipped questions and correctly matched solutions are included.
Downloaded copies are outside Tejas access controls.
Attribution: Tejas — owner-supplied paper; attribution subject to rights review

Frozen profile: 9b1d005ac164a40cb4536c886f956455dd40717d6bbd22c70bc96d87571f691d

Question 1 · Mathematics

Your response: 2 | Accepted answer(s): 2
Marks: 4 | Correct | Reason: matched_accepted_key

Review note: Internal test fixture; not academic approval. Original candidate remains in review.

Question

Q1. Let the foci of a hyperbola H be $S(-1,3)$ and $S'(-1,-1)$, and let H pass through the point $P(11,8)$. If l is the length of the latus rectum of H and d is the distance of P from the directrix of H corresponding to the focus S, then $l+2d$ is equal to :

(1) 16 **(2)** 19 **(3)** 21 **(4)** 23

Source a3721410-1000-46a2-a590-c7433fe87753 · page 1

Matched solution

Q1. Answer: (2) Hyperbola — Medium

Stem: Let the foci of a hyperbola H be $S(-1,3)$ and $S'(-1,-1)$, and let H pass through the point $P(11,8)$. If l is the length of the latus rectum of H and d is the distance of P from the directrix of H corresponding to the focus S, then $l+2d$ is equal to :

1. Dono foci line $x=-1$ par hain, so transverse axis vertical hai; centre = midpoint $(-1,1)$ aur $SS'=4 \Rightarrow c=2$.
2. $SP=\sqrt{12^2+5^2}=13$, $S'P=\sqrt{12^2+9^2}=15$; hyperbola par $|SP-S'P|=2a \Rightarrow a=1$.
3. $b^2=c^2-a^2=3$, $e=c/a=2$, latus rectum $l=2b^2/a=6$.
4. Focus $S(-1,3)$ (centre ke upar) ki directrix $y=1+a/e=3/2$; $d=8-3/2=13/2$ (check: $SP=e\,d=13$).
5. $l+2d=6+13=19$.

Final Answer: (2) 19

Trap: (1) 16 = latus rectum ko b^2/a (aadha) le liya; (3) 21 = doosre focus S' wali directrix se distance 15/2 use ki; (4) 23 = ellipse wala $b^2=a^2+c^2$ laga kar $l=10$ nikala.

Concept to recognise: Two foci on one vertical line plus a point on the curve: a comes from |SP - S'P| = 2a (vertical transverse axis), then b^2 = c^2 - a^2; 'directrix corresponding to S' signals the focus-directrix relation SP = e.d. *(Hyperbola parameter bookkeeping)*

Source 7ea0effe-ce19-4dfa-9f42-0ca8600b757c · page 1

Question 2 · Mathematics

Your response: 4 | Accepted answer(s): 4
Marks: 4 | Correct | Reason: matched_accepted_key

Review note: Internal test fixture; not academic approval. Original candidate remains in review.

Question

Q2. The sum of the series $\frac{1^3}{2} + \frac{1^3+2^3}{2+4} + \frac{1^3+2^3+3^3}{2+4+6} + \cdots + \frac{1^3+2^3+\cdots+15^3}{2+4+\cdots+30}$ is equal to :

(1) 280 **(2)** 408 **(3)** 680 **(4)** 340

Source a3721410-1000-46a2-a590-c7433fe87753 · page 1

Matched solution

Q2. Answer: (4) Sequences & Series — Medium

Stem: The sum of the series $\frac{1^3}{2} + \frac{1^3+2^3}{2+4} + \frac{1^3+2^3+3^3}{2+4+6} + \cdots + \frac{1^3+2^3+\cdots+15^3}{2+4+\cdots+30}$ is equal to :

1. n-th term: $T_n = \dfrac{1^3+2^3+\cdots+n^3}{2+4+\cdots+2n} = \dfrac{n^2(n+1)^2/4}{n(n+1)} = \dfrac{n(n+1)}{4}$.
2. Last term ka denominator $2+4+\cdots+30$ hai, i.e. $n=15$; so 15 terms hain (30 nahi).
3. $S = \dfrac{1}{4}\sum_{n=1}^{15}(n^2+n) = \dfrac{1}{4}\left(\dfrac{15\cdot16\cdot31}{6} + \dfrac{15\cdot16}{2}\right) = \dfrac{1}{4}(1240+120) = 340.$
4. Check: $\sum n(n+1)/4 = n(n+1)(n+2)/12 = 15\cdot16\cdot17/12 = 340.$

Final Answer: (4) 340

Trap: (1) 280 = sirf 14 terms tak joda (last term ka n galat, limit off-by-one); (2) 408 = 16 terms le liye; (3) 680 = denominator ko $1+2+\cdots+n$ samajh kar $T_n = n(n+1)/2$ liya (double value).

Concept to recognise: Each term is (sum of cubes)/(arithmetic sum): reduce the general term with sigma n^3 and the even-number sum to a polynomial in n, then apply sigma n^2, sigma n; the last written term fixes how many terms there are. *(Series sum via sigma formulas)*

Source 7ea0effe-ce19-4dfa-9f42-0ca8600b757c · page 2

Question 3 · Mathematics

Your response: 1 | Accepted answer(s): 1
Marks: 4 | Correct | Reason: matched_accepted_key

Review note: Internal test fixture; not academic approval. Original candidate remains in review.

Question

Q3. Let $\vec{a} = 4\hat{i} + \hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. Let $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, with $c_1c_2c_3 \neq 0$, be a vector such that $\vec{a} \times \vec{c} = 2(\vec{c} \times \vec{b})$. If the circle $x^2 + y^2 + c_1x + c_2y + c_3 = 0$ touches the y-axis and meets the x-axis at the points A and B, then the length AB is equal to :

(1) $6\sqrt{3}$ **(2)** $2\sqrt{3}$ **(3)** $6\sqrt{15}$ **(4)** $8\sqrt{15}$

Source a3721410-1000-46a2-a590-c7433fe87753 · page 1

Matched solution

Q3. Answer: (1) Vectors + Circle — Hard

Stem: Let $\vec{a} = 4\hat{i} + \hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. Let $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, with $c_1c_2c_3 \neq 0$, be a vector such that $\vec{a} \times \vec{c} = 2(\vec{c} \times \vec{b})$. If the circle $x^2 + y^2 + c_1x + c_2y + c_3 = 0$ touches the y-axis and meets the x-axis at the points A and B, then the length AB is equal to :

1. $\vec{c} \times \vec{b} = -\vec{b} \times \vec{c}$, so relation banti hai $\vec{a} \times \vec{c} + 2(\vec{b} \times \vec{c}) = \vec{0} \Rightarrow (\vec{a} + 2\vec{b}) \times \vec{c} = \vec{0}$.
2. Isliye $\vec{c} \parallel \vec{a} + 2\vec{b} = 4\hat{i} + 2\hat{j} + 3\hat{k}$, i.e. $\vec{c} = \lambda(4\hat{i} + 2\hat{j} + 3\hat{k})$, $(c_1, c_2, c_3) = (4\lambda, 2\lambda, 3\lambda)$.
3. Circle $x^2 + y^2 + 4\lambda x + 2\lambda y + 3\lambda = 0$: $g = 2\lambda$, $f = \lambda$, $c = 3\lambda$. y-axis ko touch kare $\Rightarrow f^2 = c \Rightarrow \lambda^2 = 3\lambda \Rightarrow \lambda = 3$ ($\lambda = 0$ reject, kyunki $c_1c_2c_3 \neq 0$).
4. Circle: $x^2 + y^2 + 12x + 6y + 9 = 0$, $g = 6$, $c = 9$; x-axis par chord $AB = 2\sqrt{g^2 - c} = 2\sqrt{36 - 9} = 6\sqrt{3}$.

Final Answer: (1) $6\sqrt{3}$

Trap: (2) $2\sqrt{3} = \vec{c} \times \vec{b}$ ko $\vec{b} \times \vec{c}$ padh kar $(\vec{a} - 2\vec{b}) \times \vec{c} = 0$ liya, jisse $\vec{c} = (-4, 2, 1)$ aur chord $2\sqrt{4 - 1}$; (3) $6\sqrt{15}$ = chord formula mein $g = c_1 = 12$ rakh diya ($c_1/2$ nahi); (4) $8\sqrt{15}$ = factor 2 drop karke $(\vec{a} + \vec{b}) \times \vec{c} = 0$ liya ($\vec{c} = (32, 8, 16)$).

Concept to recognise: One cross-product relation between an unknown vector c and two known vectors: move both terms to one side (watch the order c x b = -b x c), factor as (combination) x c = 0, so c is parallel to the combination; the circle built from c's components supplies the condition that fixes the scalar. *(Cross-product parallel or coplanarity condition)*

Source 7ea0effe-ce19-4dfa-9f42-0ca8600b757c · page 2

Question 4 · Mathematics

Your response: 3 | Accepted answer(s): 3
Marks: 4 | Correct | Reason: matched_accepted_key

Review note: Internal test fixture; not academic approval. Original candidate remains in review.

Question

Q4. Let $\vec{d}$ be a vector such that $\vec{d} \times (\hat{i} + \hat{j} - \hat{k}) = -\hat{i} + \hat{j}$ and $\vec{d} \cdot \hat{k} = -1$. A line L, parallel to $\vec{d}$, meets the lines $L_1 : \frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $L_2 : \frac{x}{-2} = \frac{y-3}{1} = \frac{z+3}{2}$ at the points P and Q respectively. Then the length PQ is equal to :

(1) 4 **(2)** 6 **(3)** 9 **(4)** 3

Source a3721410-1000-46a2-a590-c7433fe87753 · page 1

Matched solution

Q4. Answer: (3) 3D Geometry + Vectors — Medium

Stem: Let $\vec{d}$ be a vector such that $\vec{d} \times (\hat{i} + \hat{j} - \hat{k}) = -\hat{i} + \hat{j}$ and $\vec{d} \cdot \hat{k} = -1$. A line L, parallel to $\vec{d}$, meets the lines $L_1 : \frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $L_2 : \frac{x}{-2} = \frac{y-3}{1} = \frac{z+3}{2}$ at the points P and Q respectively. Then the length PQ is equal to :

1. $\vec{d} = p\hat{i} + q\hat{j} + r\hat{k}$: $\vec{d} \times (\hat{i} + \hat{j} - \hat{k}) = (-q - r)\hat{i} + (p + r)\hat{j} + (p - q)\hat{k} = -\hat{i} + \hat{j} \Rightarrow p = q,\ q + r = 1$; $\vec{d} \cdot \hat{k} = r = -1 \Rightarrow p = q = 2$, i.e. $\vec{d} = 2\hat{i} + 2\hat{j} - \hat{k}$ (ya $\vec{d} = \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2} + \mu\vec{a}$ se).
2. $P(2s, 2s, s)$ on L_1, $Q(-2t, 3 + t, -3 + 2t)$ on L_2; $\vec{PQ} = (-2t - 2s,\ 3 + t - 2s,\ -3 + 2t - s) \parallel (2, 2, -1)$.
3. Pehle do components barabar: $-2t - 2s = 3 + t - 2s \Rightarrow t = -1$; phir $\frac{2-2s}{2} = \frac{-5-s}{-1} \Rightarrow 1 - s = 5 + s \Rightarrow s = -2$.
4. $P(-4, -4, -2)$, $Q(2, 2, -5)$, $\vec{PQ} = (6, 6, -3) = 3\vec{d}$, so $PQ = 3|\vec{d}| = 9$.

Final Answer: (3) 9

Trap: (1) $4 = L_1, L_2$ ki shortest distance nikaal di, di hui direction ignore karke; (2) 6 = cross product ka order ulta, $(\hat{i} + \hat{j} - \hat{k}) \times \vec{d} = -\hat{i} + \hat{j}$ solve kiya, jisse $\vec{d} = -\hat{k}$ aur $PQ = 6$; (4) 3 = sirf parameter k (ya $|\vec{d}|$) report kiya, $PQ = 3|\vec{d}|$ nahi liya.

Concept to recognise: A line whose direction is defined by a cross-product equation (d x a = b plus a dot condition) 'meets' two given lines: solve the vector equation for d first (componentwise, or d = (a x b)/|a|^2 + mu a), then take general points on both lines, make PQ proportional to d and solve; the length PQ is asked, not the shortest distance. *(Line with given direction meeting two lines)*

Source 7ea0effe-ce19-4dfa-9f42-0ca8600b757c · page 3

Question 5 · Mathematics

Your response: 2 | Accepted answer(s): 2
Marks: 4 | Correct | Reason: matched_accepted_key

Review note: Internal test fixture; not academic approval. Original candidate remains in review.

Question

Q5. Let $L_1 : 3x - 4y + 5 = 0$ and $L_2 : 5x + 12y - 29 = 0$ be two lines. If the bisector of the obtuse angle between L_1 and L_2 meets the coordinate axes at the points P and Q, then the area (in sq. units) of the triangle OPQ, O being the origin, is :

(1) $\frac{225}{16}$ **(2)** $\frac{25}{4}$ **(3)** $\frac{289}{16}$ **(4)** $\frac{9}{2}$

Source a3721410-1000-46a2-a590-c7433fe87753 · page 1

Matched solution

Q5. Answer: (2) Straight Lines — Medium

Stem: Let $L_1 : 3x - 4y + 5 = 0$ and $L_2 : 5x + 12y - 29 = 0$ be two lines. If the bisector of the obtuse angle between L_1 and L_2 meets the coordinate axes at the points P and Q, then the area (in sq. units) of the triangle OPQ, O being the origin, is :

1. Dono lines ko positive constant ke saath likho: $3x - 4y + 5 = 0$ aur $-5x - 12y + 29 = 0$. Yahan $a_1a_2 + b_1b_2 = (3)(-5) + (-4)(-12) = 33 > 0$, so '+' wala bisector obtuse angle ka bisector hai.
2. Obtuse-angle bisector: $\frac{3x - 4y + 5}{5} = \frac{-5x - 12y + 29}{13} \Rightarrow 39x - 52y + 65 = -25x - 60y + 145 \Rightarrow 8x + y - 10 = 0$. (Doosra, $\frac{3x - 4y + 5}{5} = -\frac{-5x - 12y + 29}{13}$, gives $x - 8y + 15 = 0$, the acute-angle bisector.)
3. $8x + y = 10$: $P = (5/4, 0)$, $Q = (0, 10)$.
4. Area $OPQ = \frac{1}{2} \cdot \frac{5}{4} \cdot 10 = \frac{25}{4}$.

Final Answer: (2) $\frac{25}{4}$

Trap: (1) 225/16 = sign test ignore karke acute-angle bisector $x - 8y + 15 = 0$ le liya (intercepts -15, $15/8$); (4) 9/2 = normalise kiye bina $L_1 - L_2$ $(x + y = 3)$ ko bisector maan liya; (3) 289/16 = normalise kiye bina $L_1 + L_2$ $(x + 8y = 17)$ le liya.

Concept to recognise: 'Bisector of the obtuse angle between two given lines': write both lines with positive constants, apply the sign test on a1a2 + b1b2 to pick the right sign in |L1|/sqrt(a1^2+b1^2) = +-|L2|/sqrt(a2^2+b2^2), then use that line (here its intercepts). *(Angle bisector of two lines, internal or obtuse)*

Question 5 · Matched solution · continued

Source 7ea0effe-ce19-4dfa-9f42-0ca8600b757c · page 3